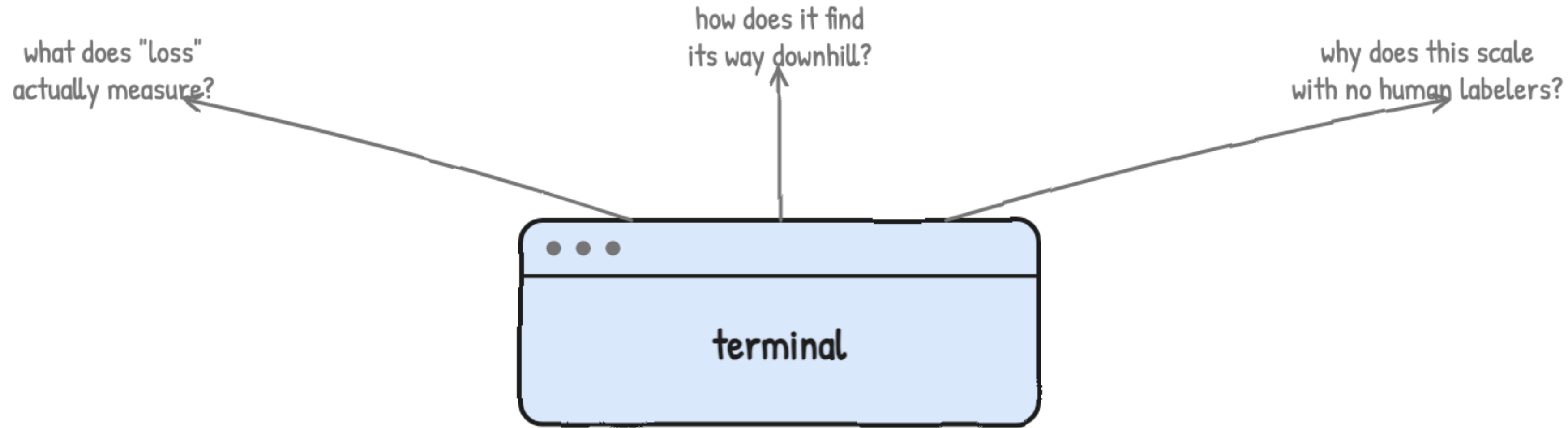


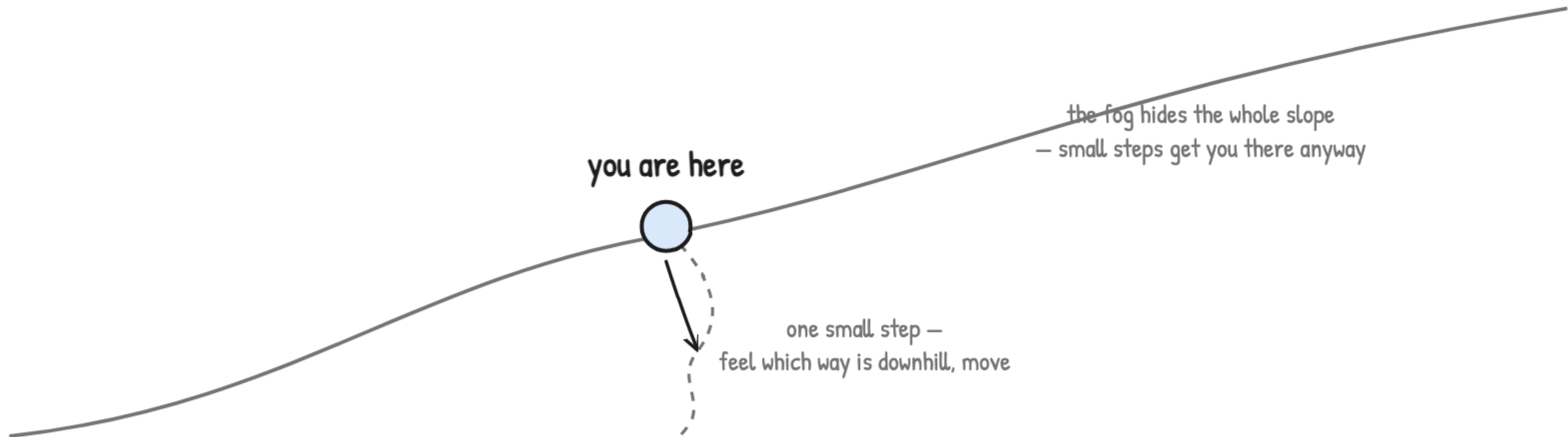
Three questions this deep dive answers



What loss, gradient, and epoch actually mean

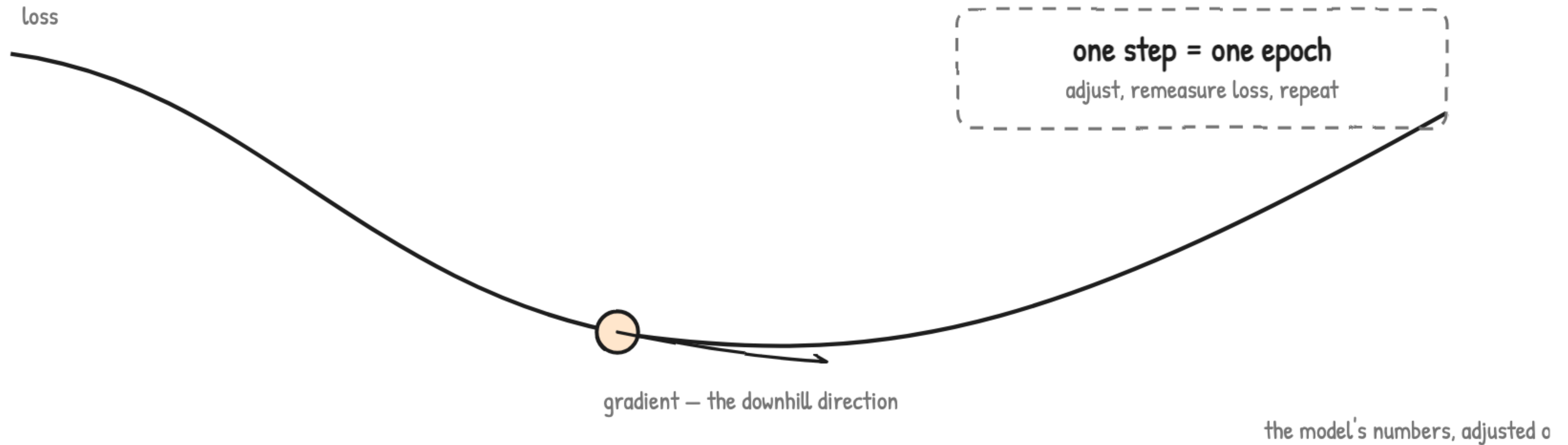
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Hiking downhill in fog, one small step at a time



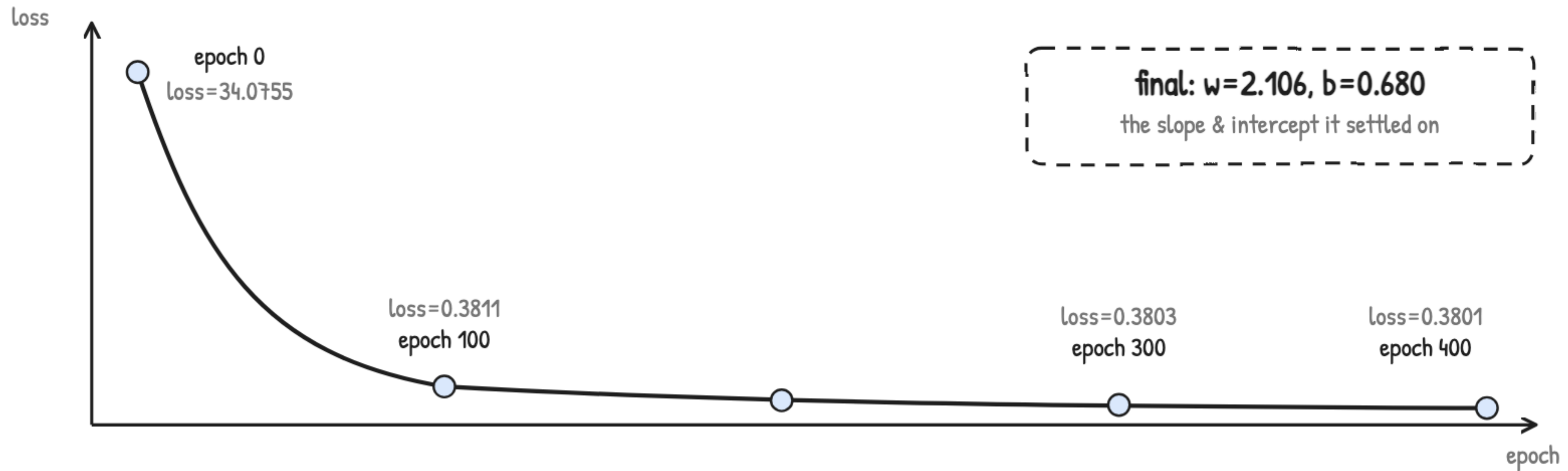
You can't see the bottom, but you can feel which way is down

Loss is the hill; the gradient is which way is downhill



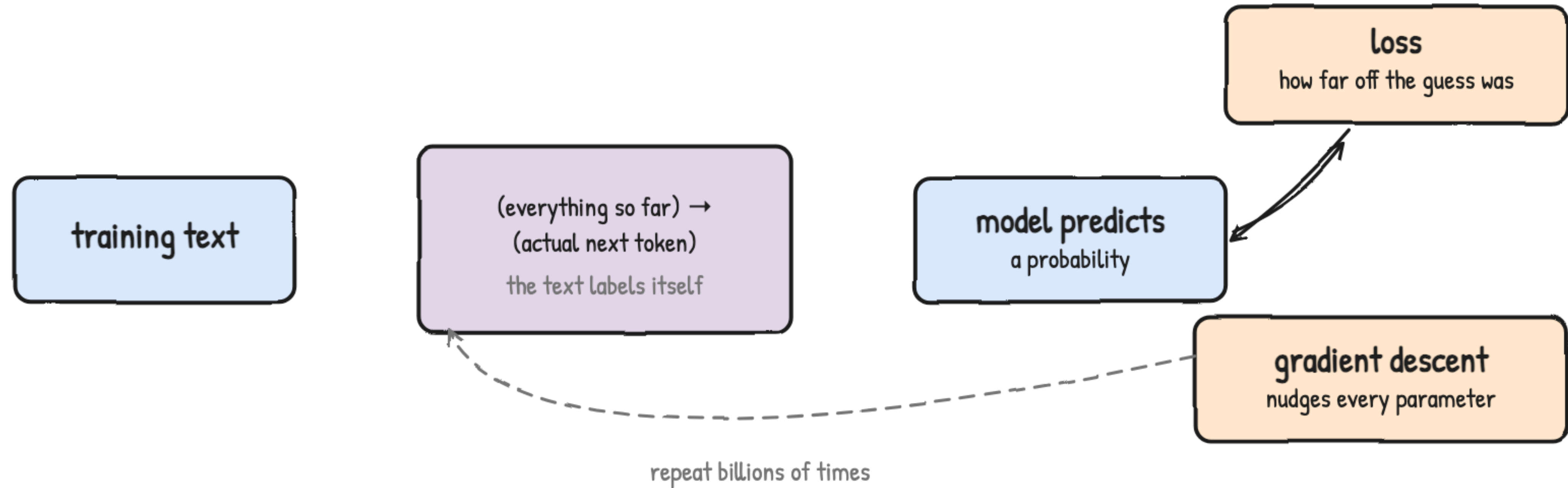
One small step is one epoch

Watch it drop fast, then crawl the rest of the way



Five points, fit with nothing but a loop

Next-token prediction is this exact loop, at internet scale



The text labels itself — no human tagger needed

Now: run the script, watch these exact numbers print

- 1 gradient descent — loss, gradient, epoch, printed live
- 2 next-token prediction — the same loop, at internet scale

```
labs/m3 — python3 gradient_descent.py
```

Same loop `model.fit()` ran for you · Gourav Shah · School of DevOps & AI